

FM Transmitter

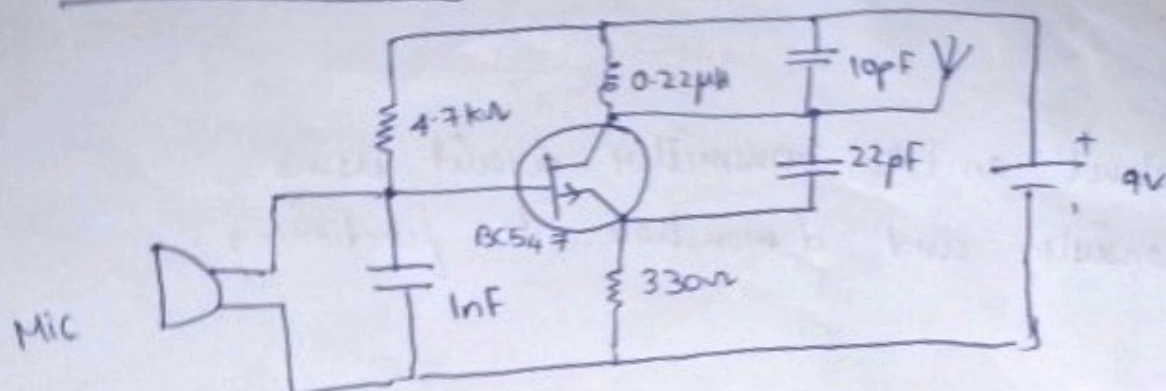
AIM:

To construct an FM transmitter circuit using BC547 transistor and demonstrate its functioning.

MATERIALS REQUIRED :

- 1) BC547 Transistor
- 2) $10\mu\text{F}$ capacitor
- 3) $22\mu\text{F}$ capacitor
- 4) 1nF capacitor
- 5) 330Ω resistor
- 6) $4.7\text{k}\Omega$ resistor
- 7) $0.22\mu\text{H}$ inductor
- 8) 9V DC power supply
- 9) Antenna wire
- 10) Microphone.

Circuit Diagram:



THEORY:

- 1) The transistor acts as an ~~oscillator~~ ^{amplifier} of carrier input signal. The $0.22\mu\text{H}$ inductor and 10pF capacitor form a tuned LC circuit. The values of L and C determine the carrier frequency.

$$f_{\text{carrier}} = \frac{1}{2\pi\sqrt{LC}}$$

FM Broadcast Band lies between $88\text{MHz} - 108\text{MHz}$.

- 2) Modulator: Input audio signals are coupled to the transistor's base by the 1nF capacitor. Variations in the input signal modulates the transistor's base current, which in turn causes variations in the carrier signal.

The 1nF capacitor removes all low frequency parts of the input audio signal.

The 22pF is required to sustain the LC circuits as it is not a perfect LC circuit (it will have some resistance).

~~Ques~~ Current Gain $= \alpha = \frac{I_c}{I_B}$
(In CB configuration)

The 330 Ω resistor maintains V_E which ensures that there is voltage gain.

$$A = \frac{\beta R_c}{R_B}, \quad \beta = \frac{\alpha}{1-\alpha}$$

Antenna: The modulated signal from the transistor is broadcasted with the help of the antenna.

OBSERVATIONS:

$$f_{\text{carrier}} = \frac{1}{2\pi \sqrt{0.22\mu \cdot 10p}} = 107.3 \text{ MHz}$$

Using a longer antenna provides better clarity

RESULT:

Hence, the FM transmitter has been built and techniques of signal processing and transmission have been used for the same.